SYSTEM DOCUMENTATION

Al Qamar Hospital Management System

Modules, user roles, workflows, business rules, patient lifecycle, admissions, medical records, laboratory, medical store and inventory, and administrative and financial management

VERSION	1.0 — reflects the deployed application as of 21 October 2026 (source commit `e015a8c`)
PREPARED FOR	Al Qamar Hospital management, department leads and the development/maintenance team
BASIS	Derived directly from the application source code, database schema and in-repository design notes
CURRENCY / TIME	Pakistani Rupee (PKR) · Clinic day in Asia/Karachi (PKT, UTC+5)

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1 About this document

1.1 Purpose

This document describes the Al Qamar Hospital Management System (HMS) as it is built and deployed today. It explains what each module does, who uses it, the step-by-step workflows staff follow, and the business rules the software enforces. It is intended to serve as:

- a **reference for management** on how money, patients and stock move through the hospital;
- a **training aid** for front-desk, laboratory, pharmacy and management staff;
- a **functional specification of the system as built**, for developers maintaining or extending it, and for auditors.

1.2 Scope and source

Everything in this document was derived from the application itself: its database schema, server-side business logic, API route handlers and screens. Where the software enforces a rule, the rule is stated as the system behaves. It is not a statement of intended policy. Section 18 records observations and gaps found during the analysis.

1.3 Conventions

Convention	Meaning
PKR	All money is in Pakistani Rupees, stored to 2 decimal places (medicine unit costs to 4).
"Today", "the day"	The clinic's calendar day in Asia/Karachi. The server is set to this time zone at start-up.
Front desk / Operator	Staff with the `OPERATOR` role.
Store	The in-house pharmacy ("medical store"), staff with the `MEDICAL_STORE` role.
BR-nn	A numbered business rule, collected in Section 14.
`/path`	A screen (page address) in the application.

2 Executive summary

The Al Qamar HMS is a web application used all day by staff on the hospital network. It runs the front desk, the laboratory, the operation theatre, in-patient admissions and the in-house pharmacy, and it keeps one shared cash ledger that management uses for revenue, doctor payouts, salaries and expenses.

Clinical and front-desk operations

Patient registration with MR numbers; OPD consultations with per-doctor daily tokens and bilingual (English/Urdu) slips; diagnostic billing for Laboratory, X-Ray, Ultrasound, ECG and ECO; minor emergency visits; operation theatre cases; full in-patient admissions.

Laboratory and records

A test and price catalog with structured result templates (around 60 ready-made ones), typed result entry with automatic High/Low flags, 80 mm printable reports, live "result ready" notifications to the desk, and a per-patient history.

Pharmacy and stock

Batch- and expiry-tracked medicine stock with packaging levels, first-expiry-first-out sales, in-patient credit, returns, adjustments, physical stock counts, a complete stock ledger, supplier balances and an end-of-day cash closing.

Key capabilities at a glance

Area	What the system provides
One money ledger	Every rupee received anywhere is one *Payment* row that belongs to exactly one source (OPD, test receipt, OT, emergency, medicine sale or admission). Dashboards and exports read this single ledger, so departments cannot double-count.
Doctor accounting	Each doctor's share is computed automatically from the OPD split, OT surgeon fees, admission visit fees and diagnostic referral percentages. A running balance shows what the hospital owes each doctor.
In-patient billing	Room, doctor-visit and technician charges accrue per calendar day without any scheduler. The bill keeps the hospital part and the pharmacy part separate, and discharge is blocked until both are settled.
Controls	Discounts, free consultations, cancellations and refunds require a reason. Every change is audit-logged. Numbering is concurrency-safe. Stock can never go negative and expired stock can never be sold.
Reporting	A management dashboard with gross collection, doctor share, hospital share, expenses and net revenue; Excel exports for billing and the store; and separate dashboards for the front desk, lab and store.

Users

Four roles, each landing on its own workspace: `OPERATOR` (front desk), `LAB`, `MEDICAL_STORE` and `MANAGEMENT`. Accounts are created by management. They are never deleted, only deactivated, and every new or reset account must choose its own password before it can do anything.

3 System overview and architecture

3.1 Module map

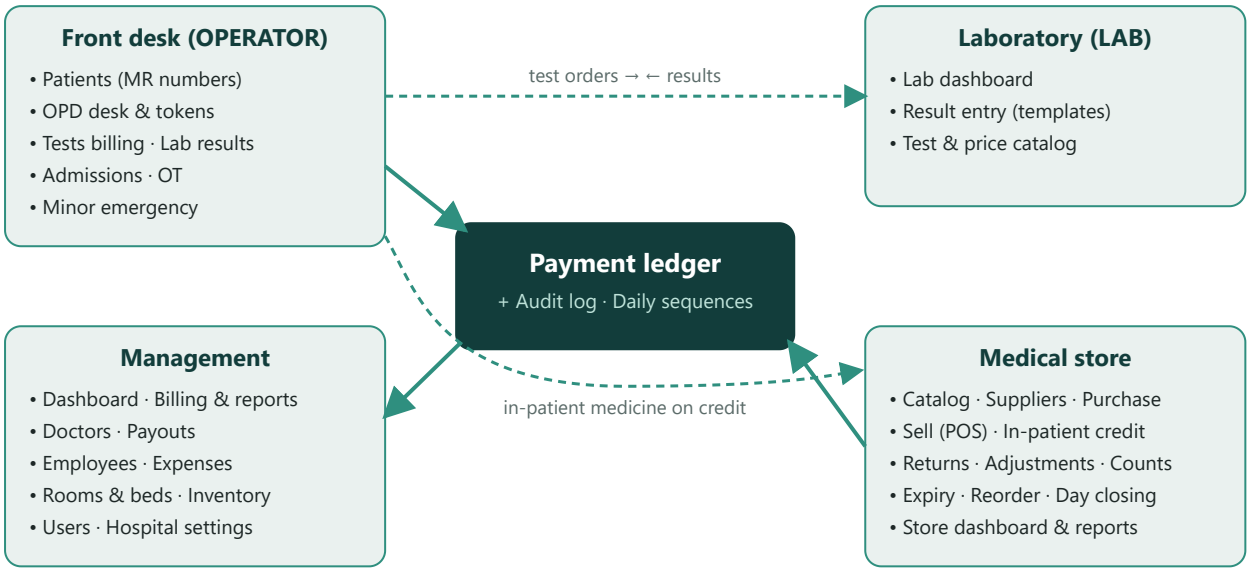


Figure 3.1 — Modules and how they connect. All money flows into one Payment ledger that management reports on.

3.2 Architecture

Staff browsers
Front desk, lab, store and management PCs on the hospital network. Thermal 80 mm printers for slips and receipts.

→

Web application
Next.js (React) pages plus server API routes. Each API checks the session and the role before doing anything.

→

Business rules
Shared server modules: admission billing, doctor shares, payment sources, salary cycle, packaging, lab templates.

→

MySQL / MariaDB
About 40 tables on InnoDB, accessed through Prisma. Transactions keep multi-step saves atomic.

The application is a single web server process with a relational database. There is no background scheduler: anything time-based (such as daily in-patient charges) is calculated when a record is opened or listed. Live screens (lab queue, results desk, dashboards) refresh themselves by polling, pause when the browser tab is hidden, and slow down after failures.

3.3 Technology stack

Layer	Technology
Framework	Next.js 16 (App Router), React 19, TypeScript 5 (strict)
User interface	Chakra UI v3 with Emotion, Tailwind v4 (global styles), Font Awesome icons, react-toastify messages, Recharts charts
Data	Prisma 7 ORM with the MariaDB driver adapter, MySQL/MariaDB on InnoDB
Exports	ExcelJS (management billing report and store report); printable HTML for PDF/thermal printing
Hosting	GoDaddy cPanel "Setup Node.js App" (production); WAMP MySQL on developer PCs

3.4 Design principles visible in the system

1. **One source of truth for money.** Each payment belongs to exactly one source, and a single shared helper labels it, so a payment can never be counted under two departments.
2. **Snapshots, not live links, for anything printed or billed.** Test names and prices are copied onto receipts at sale. Result templates are copied into each report. Room, doctor and technician rates are copied into each admission and each charged day. Later changes to the catalog never rewrite history.
3. **Reasons for every exception.** Discounts, free consultations, cancellations, refunds, adjustments and admission cancellations all require a written reason, and the reason is stored in the audit log.
4. **Safe numbering.** Tokens and document numbers come from a daily-sequence table inside a database transaction. The transaction is retried automatically if two users save at the same instant.
5. **No scheduler dependency.** Daily admission charges are added idempotently whenever a stay is viewed or changed, so each day is charged exactly once even on shared hosting.

4 User roles and access control

4.1 Roles

Role	Who	Responsibilities
OPERATOR	Front-desk / reception staff	Registers patients, issues OPD slips and tokens, bills diagnostic tests, hands out lab results, records emergency visits, runs OT cases and in-patient admissions (payments, invoices, discharge).
LAB	Laboratory technicians	Maintains the test and price catalog and result templates, and enters results for Laboratory tests.
MEDICAL_STORE	Pharmacy staff	Runs the whole medical store: catalog, suppliers, purchases, sales, in-patient credit, returns, stock control, day closing and store reports.
MANAGEMENT	Owners / administrators	Sees the financial dashboard and reports, manages doctors, payouts, employees, expenses, rooms and beds, general inventory, user accounts and hospital settings. Can also call most other roles' functions.

After sign-in every role lands on `/`. Management and operators see their own dashboards there. LAB users are redirected to `/lab` and MEDICAL_STORE users to `/medical-store`.

4.2 Navigation per role

Role	Sidebar pages
OPERATOR	Dashboard · Patients · OPD desk · Tests · Lab results · Admissions · Operation theater · Minor emergency
MANAGEMENT	Dashboard · Doctors · Billing & reports · Payouts · Employees · Expenses · Hospital settings · Inventory · Rooms & beds · User access
LAB	Lab dashboard · Upload tests (result entry) · Test & Price Catalog
MEDICAL_STORE	Store dashboard · Medicine catalog · Suppliers · Supplier payments · Purchase (stock in) · Reorder list · Sell medicine · In-patients · Day closing · Stock adjustments · Stock count · Stock history · Expiry tracker · Sale returns · Purchase returns · Reports
All roles	Profile · Change password · Sign out

4.3 Permission matrix (server-enforced)

The sidebar only hides links. The real control is on the server: every API call checks that the user is signed in and that their role is on that function's allowed list, and refuses it otherwise.

Function	OPERATOR	LAB	STORE	MGMT
Register / edit patients	✓	—	—	✓
Search / view patients	✓	—	✓	✓
OPD visits, emergency, OT cases, admissions	✓	—	—	✓
Bill diagnostic tests; cancel/refund receipts	✓	—	—	✓

Function	OPERATOR	LAB	STORE	MGMT
View test results	✓	✓	—	✓
Enter results (Laboratory module)	—	✓	—	✓
Enter results (X-Ray, Ultrasound, ECG, ECO)	—	—	—	✓
Test & price catalog (create/edit)	—	✓	—	—
All medical store functions	—	—	✓	✓
View admission technician fees	✓	—	—	✓
Doctors, payouts, employees, expenses, reports	—	—	—	✓
Rooms & beds, admission fees, general inventory	—	—	—	✓
User accounts, hospital settings	—	—	—	✓

Note: only the LAB role can edit the test catalog; management cannot. Store medicine pricing is maintained by the store, not management.

4.4 Authentication and account rules

Rule	Behaviour
Passwords	At least 8 characters. Stored as salted scrypt hashes, never in plain text.
Sessions	A random token in a secure cookie; the database keeps only its SHA-256 hash. A session expires after **8 hours**. HTTPS is mandatory in production.
Login throttling	5 wrong passwords for one username lock that username for **15 minutes**. The lock is per username, not per IP, because all clinic PCs share one public IP.
Forced password change	Every newly created or reset account is flagged "must change password". The user cannot use any page until they set their own password.
Password reset	Management resets a password to a random one-time password, shown once on screen. All of that user's sessions are signed out.
Voluntary change	From the profile menu, the user must re-enter their current password.
No deletion	Users are never deleted or renamed, only deactivated, which also ends their sessions. A manager cannot deactivate their own account.
Usernames	3–80 characters: lowercase letters, digits, dot, underscore and hyphen.
Developer console	`/dev` (backup, restore, wipe database, create a management login) uses its own password and signed cookie, and is **switched off in production** unless deliberately enabled.

5 The patient lifecycle

A patient is registered once and receives a permanent MR number. Every later encounter — consultation, test, emergency, operation, admission or pharmacy purchase — is linked to that one record.

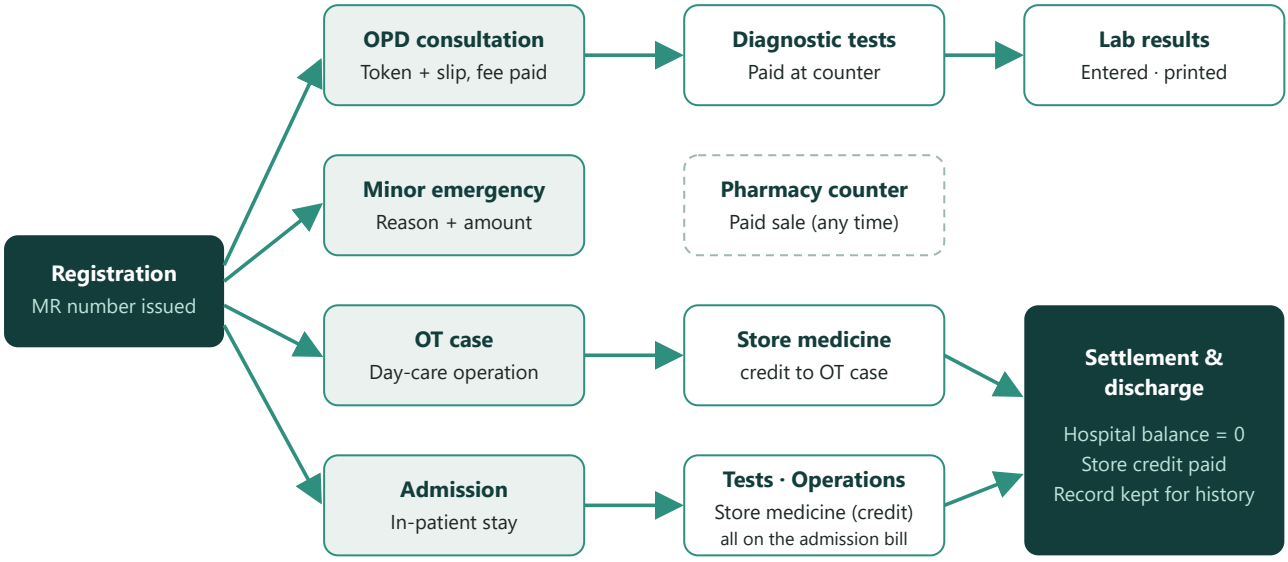


Figure 5.1 — Patient journeys. OPD, emergency and counter pharmacy are paid on the spot. OT cases and admissions run a balance that must be cleared before discharge.

5.1 Encounter types compared

Encounter	Started by	Payment	Ends when
OPD visit	Operator, OPD desk	Fee paid when the slip is issued (or waived as FREE with a reason)	Immediately: a one-off visit
Diagnostic receipt	Operator (against an OPD number) or from an admission	OPD: paid at the counter. Admission: added to the in-patient bill	When the results are entered and handed over
Emergency visit	Operator	Amount paid on the spot	Immediately
OT case	Operator, OT page	Full, advance, or Due; instalments allowed	Discharge, only when the balance is 0 and store medicine is paid
Admission	Operator, Admissions page	Advances and part-payments; final settlement after the invoice	Discharge, only when the hospital balance is exactly 0 and store credit is settled
Counter sale	Store, Sell page	Paid on the spot	Immediately (returns possible later)

6 Front desk: registration, OPD, tests, emergency

6.1 Patient registration and records — `/patients`

Data captured

Field	Required	Rules
Name	Yes	Free text
Father's name	Yes	Free text
Gender	Yes	Selected
CNIC	No	Auto-formatted to `12345-1234567-1`; must be unique across patients
Mobile	No	Auto-formatted to `03XX-XXXXXXX` (Pakistani mobile); must be unique across patients
Date of birth	No	Valid date; age is shown in years, months or days
Address	No	Free text

- **MR number** is issued automatically as `MR-<year>-<6-digit id>`, e.g. `MR-2026-000412`. It never changes.
- **Duplicate prevention:** registering or editing a patient with a CNIC or mobile that belongs to someone else is refused, and the existing MR number is shown. The screen also checks the CNIC live while typing.
- **Search** matches any part of the MR number, CNIC, mobile or name (25 most recent matches).
- **Patient history** shows every OPD visit (doctor, fee, payment), every test receipt with its result status, and the audit trail of edits (who changed what and when).
- **Newborns** admitted to NICU are registered from the Admissions page itself (Section 8.2). They share the mother's mobile, so the mobile duplicate check is skipped for them.

6.2 OPD desk — `/opd`

1. Find patient Type-ahead search by name or MR. Register a new patient if not found. →

2. Choose doctor Search by name or speciality. Each doctor's weekly schedule shows whether they are on duty now. →

3. Payment Cash, card, bank transfer, online, or FREE. FREE needs a reason. →

4. Issue Token and OPD number assigned; fee recorded as a payment. →

5. Print Bilingual doctor slip, plus thermal doctor and patient copies.

Item	Rule
Daily token	Per doctor, per day, starting at 1. Two staff issuing at the same moment still get consecutive tokens.
OPD number	`OPD-YYYYMMDD-DDD-TTTT`: date, doctor id and token.
Fee	The doctor's current consultation fee, copied onto the visit. A FREE visit is recorded at PKR 0, method FREE, with the reason as the payment note.
Doctor availability	Configured as weekdays plus a from/to time. Used to show who is on duty. It does not block issuing a slip.

Item	Rule
Doctor slip	A full-page consultation slip in English and Urdu (doctor name, speciality and qualifications in both languages, hospital name and address in Urdu). It has a large blank area where the doctor writes the prescription by hand.
Thermal copies	Separate 80 mm "Doctor Copy" and "Patient Copy" receipts.
Deleting a visit	Needs a reason. The operator records which slips were returned (main slip, doctor receipt, patient receipt). The visit and its payment are removed and the full record is kept in the audit log. **Not allowed** once any test receipt was issued under that OPD number.

6.3 Diagnostic tests billing — `/diagnostics`

Tests are billed by the front desk against an OPD visit, so every counter test has a referring doctor (the OPD doctor).

- **Modules:** LABORATORY, X-RAY, ULTRASOUND, ECG, ECO. One receipt holds tests from one module.
- **Selecting the visit:** search by OPD number, patient name or MR number; today's OPD visits are offered as a quick list.
- **Pricing:** each test's catalog price is copied onto the receipt line (name and price at sale), so later price changes don't alter old receipts.
- **Discount:** optional, in PKR, cannot exceed the subtotal, and **requires a reason**. Discounted receipts are audit-logged as DISCOUNT.
- **Numbering:** `LAB-YYYYMMDD-0001` (prefix is the first three letters of the module: LAB, X-R, ULT, ECG, ECO). One sequence per module per day, shared with tests ordered from admissions.
- **Cancel / refund:** needs a reason. The receipt and its payment are marked CANCELLED or REFUNDED, so they drop out of income. A test billed on an admission can only be cancelled while that admission is still open (status ADMITTED).
- **Referral commission:** the referring doctor earns their per-module share percentage of the receipt total (Section 12.3).

6.4 Lab results desk — `/results`

- Lists test receipts with their result status (pending / done) and refreshes automatically every 15 seconds.
- While an operator is signed in, the workspace checks for newly completed lab results and shows a notification as soon as the lab finishes a test.
- Receipts ordered for in-patients are marked with their admission.
- **Printing:** the report reproduces the clinic's traditional result form: patient block, then *Test* | *Result* | *Units* | *Reference Range* per test, section headings and the remarks line. Printing is recorded (printed at / by).

6.5 Minor emergency — `/emergency`

- Select a registered patient, enter the reason, the amount (zero or more) and the payment method. The payment is recorded immediately.
- Gets a daily token and a visit number `EMG-YYYYMMDD-0001`, and a printable receipt.
- Income is reported as "Minor emergency".

6.6 Operator dashboard — `/`

A today-only view for the front desk, refreshed automatically. It deliberately excludes doctor shares, expenses and net revenue, which are management figures.

Counts and queues

OPD patients and the last token issued per doctor; doctors seen today and doctors scheduled but not yet seen; emergency visits; test receipts per module and how many results are back; admitted today, discharged today, currently admitted, invoiced awaiting discharge; open OT cases (booked, in progress, ready for discharge).

Money and activity

Collection at the desk today, by service and by payment method, plus the part taken by the signed-in user. Medicine is excluded because it is paid at the store's own counter. Also a patient-flow-by-hour chart and a running log of the day's registrations and payments.

7 Operation Theatre (OT) — /ot

The OT page manages operation cases. A case is either **standalone** (a day-care or short-stay operation with its own bill) or **linked to an admission**, in which case it is billed, paid and discharged only through that admission.

7.1 Creating a standalone OT case

Field	Notes
Patient, doctor	Searched like the OPD desk. The doctor must be active.
Room / bed	Typed in (configured beds are offered as suggestions only).
Procedure, diagnosis, date/time, recommendations	Clinical description of the case.
Fees	Surgeon (doctor) fee, theatre fee, anaesthesia fee, room fee, hospital fee. Total = sum of fees.
Payment method	**Must be chosen explicitly**: Cash, Card, Bank transfer, Online, or **Due** (nothing paid yet). There is no default, so a case can never silently appear paid.
Amount paid now	Blank = the whole bill. A smaller amount is an advance; the rest stays as the case balance. It cannot exceed the bill.

The case number is `OT-YYYYMMDD-0001`. A printable admission slip and invoice are offered immediately.

> **Medicine for OT patients**
> New OT cases no longer have typed-in OT or home medicine fees. The medical store issues the medicine to the OT case on credit (Section 10.4), and the patient pays the store directly. Older cases keep their historical medicine fees.

7.2 During the case

- **Add consumables and medicines (line items):** description, category, quantity, unit price, and how it is paid (a method, or Due). Adding the first item moves the case to IN_PROGRESS. Items can only be added while the case is BOOKED or IN_PROGRESS.
- **Record payment:** collects all or part of the outstanding balance, in any number of instalments. Each instalment is its own payment and counts as income on the day it is received. Overpayment is refused.
- **View invoice:** always the complete bill. Printing the slip records when it was last invoiced.
- **Ready for discharge:** marks the case clinically complete. The store then sees it as ready to settle.

7.3 OT status lifecycle

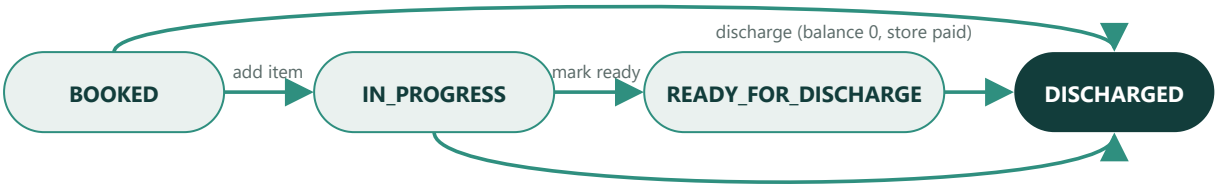


Figure 7.1 — OT case statuses. Discharge is possible from any open status once nothing is owed.

7.4 Discharge rules

An OT patient can be discharged only when:

1. the case is BOOKED, IN_PROGRESS or READY_FOR_DISCHARGE; and
2. the outstanding balance (fees + line items − payments received) is **zero**; and
3. the medical store has marked all medicine issued to this case as **paid**.

At discharge, the final total is recalculated and the discharge date is stamped. Credit sales whose medicine was fully returned are closed automatically. The surgeon's fee counts toward the doctor's earnings from the discharge date.

7.5 OT cases inside an admission

When an operation is added from an admission (Section 8.4), the case carries the admission link and:

- its room and medicine parts are zero (the room is charged daily on the admission; medicine comes from the store on credit to the admission);
- payments, "ready for discharge" and discharge are **refused on the OT page**. They happen on the admission;
- extra items may be added on the OT page **only as Due**, so they are collected on the admission bill;
- it is discharged automatically when the admission is discharged.

8 Admissions (in-patient department)

The Admissions module (`/admissions` and `/admissions/[id]`) manages in-patient stays from admission to discharge: bed allocation, daily charges, doctor visits, extra charges, operations, tests, pharmacy credit, advances, invoicing and final settlement.

8.1 Admission categories and bed rules

Admission category	Meaning	May use configured beds of category
MEDICAL	Medical treatment	Private room or Ward
SURGICAL	Surgical (operation)	Private room or Ward
CRITICAL	Critical / suspicious patients	Critical care only
NICU	Newborn intensive care	NICU only

- Beds are configured by management (Section 13) with a category (PRIVATE, WARD, CRITICAL, NICU), a daily rate, and whether that rate is per room or per bed.
- A bed is **occupied** from admission until discharge or cancellation, and cannot be given to a second patient or deactivated while occupied.
- A **manually entered room** (custom arrangement or price) is always allowed. It then needs a daily rate.
- A patient can have only **one active admission** at a time.

8.2 Admitting a patient

1. Patient Search an existing patient, or register a newborn (NICU only). → **2. Category** Selecting it fills the technician fee and clears a bed that category cannot use. → **3. Doctor & bed** Attending doctor; a free bed of an allowed category, or a manual room. → **4. Rates** Room, doctor-visit and technician daily rates (editable). → **5. Advance** Optional advance with payment method. Number `ADM-YYYYMMDD-0001` .

Rate	Default (copied in at admission; later master-data changes don't affect this stay)
Daily room rate	The selected bed's daily rate (required for a manual room)
Daily doctor fee	The attending doctor's "admission daily fee"
Daily technician fee	The category's technician fee set on the Rooms & beds page (hospital income; technicians are salaried)

Also recorded: diagnosis, notes, attendant name and attendant mobile (validated `03XX-XXXXXXX`). For a newborn, the baby's name (e.g. "Baby of Ayesha"), father's name and gender are required; date of birth defaults to today.

8.3 Daily charges (accrual)

Day-counting rule

A stay shorter than 24 hours is charged as **one day**. Once it reaches 24 hours, **every calendar date** from the admission date to the end date is charged. The end is now, or the invoice time once invoiced.

Example: admitted on the 1st at 10:00 and still admitted on the 2nd at 14:00 → 2 days. Admitted 1st at 22:00, invoiced 2nd at 08:00 → 1 day.

- Each charged date becomes one **day row** holding that day's room label and rate, the doctor and doctor fee, and the technician fee.
- Day rows are added **automatically whenever the admission is listed, opened or changed**. There is no scheduler. Only missing days are added, at the stay's current rates, and existing days are never changed. Each date can exist only once, so a day is never charged twice.
- **Doctor did not arrive:** while admitted, the operator can untick a day's doctor visit. That day's doctor fee is then excluded from the bill and from the doctor's earnings.
- **Changing room, doctor or rates** (UPDATE_STAY): applies from the next charged day. "Also the current day" re-prices the latest charged day too, for example to correct a wrong rate straight after admission. Days already charged keep their rates.

8.4 During the stay (while status is ADMITTED)

Action	What happens
Collect payment	Any amount is accepted as an advance while admitted. After the invoice, no more than the balance due. Notes are labelled advance / part-payment / final payment.
Add / remove extra charge	Anything else billed (oxygen, a procedure, consumables…). Description and amount; hospital income.
Add operation	Creates a linked OT case (Section 7.5) with surgeon, theatre, anaesthesia and hospital fees. Its total goes on the admission bill.
Order tests	Choose a module and tests from the same catalog as the OPD counter. A diagnostic receipt is created with no payment; its total goes on the hospital bill. The attending doctor is the referring doctor. The lab sees it in the normal queue.
Cancel test	Needs a reason. Refused once the lab has entered any result.
Store medicine	Issued by the pharmacy on credit to the admission (Section 10.4), for Ward, OT or Home use.
Mark doctor visit	Tick or untick per day (Section 8.3).
Update stay	Change bed (subject to category rules and occupancy), doctor, rates, diagnosis or notes.

8.5 The admission bill

The bill has two independent parts, kept apart so that both the hospital's and the pharmacy's books stay correct:

Hospital part (collected by the operator)

Room total (all charged days)
+ Doctor visits (only days the doctor came)
+ Technician fees (all charged days)
+ Extra charges
+ Linked operations (not cancelled)
+ Tests ordered (not cancelled)
= Hospital total
− Admission payments (incl. any negative refund row) and payments on linked operations
= Hospital balance (negative = excess advance)

Medical store part (collected by the store)

Medicine issued on credit, by purpose: Ward, OT, Home (discharge)
− Medicine returned before payment (credit returns)
= Store total

What is **owed** to the store is the total of sales still UNSETTLED. It becomes store income only when the store marks it paid.

8.6 Status lifecycle

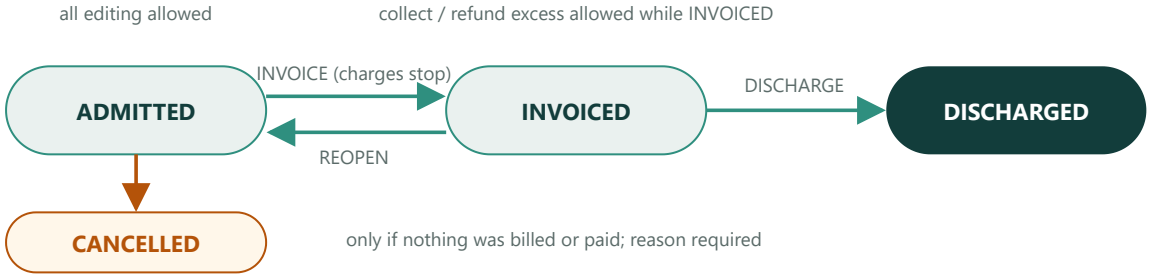


Figure 8.1 — Admission statuses.

Status	Meaning and what is allowed
ADMITTED	Patient in bed; charges accrue daily. All actions allowed. The store can issue medicine.
INVOICED	Final invoice generated; charges stop at the invoice time and the store can no longer issue medicine. Only collecting the balance, refunding an excess, reopening or discharging are allowed. The bed stays occupied.
DISCHARGED	Closed. The bed is released; linked OT cases are discharged; doctor visit fees now count toward doctor earnings.
CANCELLED	Admitted by mistake. Only possible from ADMITTED, with a reason, and only if there are no payments, medicines, active operations or active tests.

8.7 Discharge

Discharge is allowed only when all of these hold:

1. the admission is **INVOICED**;
2. the hospital balance is **exactly zero**. If the patient owes money, collect it. If the patient paid more than the bill, use **Refund excess** first, which records a negative payment on the day the money is handed back;
3. nothing is **owed to the medical store**: the store must have marked the medicine bill paid.

On discharge the system stamps the discharge time, discharges any open linked OT cases, and closes credit sales whose medicine was entirely returned. Every admission action is audit-logged with its details.

9 Medical records and laboratory

9.1 What the patient record holds

The system is an operations, billing and diagnostics system. The patient's medical record consists of:

Record	Content
Demographics	MR number, name, father's name, gender, date of birth, CNIC, mobile, address, with an audit trail of every edit.
OPD history	Every consultation: date, doctor, token, fee, payment method.
Diagnostic history	Every test receipt with its tests, typed result values, flags, reference ranges, remarks, who entered them and when.
Emergency visits	Date and reason.
OT cases	Procedure, diagnosis, procedure date/time, recommendations, items used.
Admissions	Category, diagnosis, notes, attendant, bed history (per day), attending doctor per day, doctor attendance, operations, tests and medicines issued (with purpose Ward/OT/Home).
Pharmacy history	Medicine sales linked to the patient.

> **Scope note**
> Prescriptions, vital signs, nursing notes and progress notes are **not** recorded electronically. The OPD slip has a printed blank area where the doctor writes the prescription by hand.

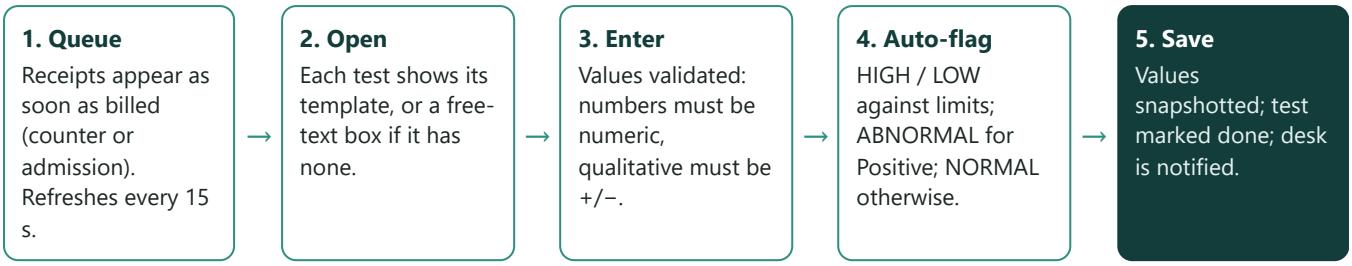
9.2 Test catalog and result templates — `/configuration` (LAB)

- Each test has a module, name (unique within the module), price, active flag, and optional **default remarks** printed under its results (e.g. "Test performed by chromatography method").
- Laboratory tests can have a **result template**: an ordered list of rows. Each row is one of:

Row kind	Purpose and behaviour
HEADING	Section title (e.g. "Differential count"); no value. Printed only if a value under it is filled in.
NUMERIC	A number with unit, optional low/high reference limits, optional printed reference text, and an optional second unit with a conversion factor (e.g. mg/dl × 0.0555 = mmol/l, kept to 3 decimals).
QUALITATIVE	Negative / Positive. The default reference is "Negative (-Ve)".
TEXT	Free text (e.g. colour, appearance).

Ready-made templates: about 60 common tests can be loaded into the editor in one click (CBC, LFT, RFT, lipid profile, thyroid profile, HbA1c, electrolytes, urine and stool R/E, dengue, typhoid, hepatitis, HIV, hormones, cardiac enzymes, cross-match, and others). They use typical adult reference ranges. The lab is expected to check them against its own analysers and kits before saving.

9.3 Result entry workflow — `/lab/upload`



- **Snapshotting:** each saved value stores its own copy of the row name, unit, alternate value and reference text. Editing a template later never changes a report already issued.
- **Corrections:** saving a test again replaces its earlier values, so no stale rows are left.
- **Done rule:** a test is done when it has free text or at least one non-heading value. A receipt is DONE when all its tests are done.
- **Who enters what:** the LAB role enters **Laboratory** results only. X-Ray, Ultrasound, ECG and ECO need the patient present, so the LAB role cannot enter them; management can.
- Legacy PDF-upload columns still exist in the database for old data but are no longer used.

9.4 Reports and notifications

- The front desk prints the report on 80 mm paper from the Lab results page (Section 6.4). In-patient test results also appear on the admission's page.
- When the lab saves results, signed-in operators get an on-screen notification.

9.5 Lab dashboard — `/lab`

Tiles: pending tests (right now), completed tests, new receipts, tests ordered and cash collected for the selected period. Charts: receipts over the last 7 days, 30 days or 12 months; result and print status; and a test-type breakdown (top 7 tests plus "Other"). Refreshes every 15 seconds.

10 Medical store (pharmacy)

The medical store is a self-contained pharmacy subsystem with its own catalog, suppliers, expiry-dated batch stock and point of sale. It shares the hospital's patient register and the central payment ledger. Access: MEDICAL_STORE and MANAGEMENT.

10.1 Medicine catalog and packaging levels

- Fields: name, generic name, category (Tablet, Syrup, Injection, Capsule, Ointment, Other), **base unit** (Strip, Bottle, Box, Piece or Vial), sale price per base unit, low-stock (reorder) level, active flag. Name + category must be unique.
- **Packaging levels** describe how a medicine is packed, each relative to the level below. For example, with base unit Strip, "1 Box = 10 Strips" and "1 Carton = 12 Boxes" means 1 Carton = 120 Strips. Levels are unlimited and per medicine. Level names must be unique and must differ from the base unit (name them BOX, CARTON, PACK…).
- All stock is held and sold in base units. Quantities are entered and displayed in any level.

10.2 Suppliers and purchasing

Function	Behaviour
Suppliers	Name, contact person, phone, address. Can be edited and deactivated or reactivated. Only active suppliers appear in purchase and return pickers.
Purchase (stock in)	Pick a supplier, then one line per medicine: batch number, packaging level received, quantity at that level, **total cost paid**, optional batch sale-price override, and expiry date. Number `PUR-YYYYMMDD-0001`.
Batches	Every purchase line creates a new, separately tracked batch. The system converts the quantity to base units and derives the cost per base unit (4 decimals). Nothing merges into existing stock.
Supplier payments	Record money paid to a supplier (amount, method, date not in the future, note). Each payment is posted as a **SUPPLIER_PAYMENT** hospital expense on that date.
Supplier balance	Owed = purchases − payments − purchase returns taken as credit.
Reorder list	Medicines at or below their low-stock level, counting only non-expired stock. Suggested quantity tops stock up to **twice** the low-stock level. Printable.

10.3 Selling — `/medical-store/sell`

1. Customer A registered patient (by MR/name) or a walk-in name. → **2. Lines** Quantity per packaging level (e.g. 2 boxes + 3 strips); live stock and price shown. → **3. Discount** Optional PKR amount; reason required; not above subtotal. → **4. Pay** Cash, card, bank transfer or online. → **5. Receipt** 80 mm receipt with daily token; `MED-YYYYMMDD-0001`.

FEFO: First-Expiry-First-Out

Each line draws from the batch that **expires soonest**, splitting across batches when one is not enough. **Expired batches are never sold.** If non-expired stock is short, the sale is refused with the available quantity. Each line's price is the batch override price if set, otherwise the catalog price. The batch cost is recorded on the sale line for profit reporting and never printed on the customer receipt.

10.4 In-patient credit — `/medical-store/inpatients`

The In-patients page lists every **currently admitted** patient and every **open standalone OT case**, with what the store has issued so far, plus every stay that still owes the store.

Step	Rule
Issue medicine	Opens the Sell page in in-patient mode: the patient is fixed, there is no discount and no payment, and the store chooses the purpose: **Ward**, **OT** or **Home (discharge)**. Stock still leaves FEFO. The sale is recorded as **UNSETTLED** credit on the patient's bill.
Issue window	Admission: only while ADMITTED (it stops at the final invoice). OT case: while BOOKED, IN_PROGRESS or READY_FOR_DISCHARGE. This is checked again inside the transaction in case the operator invoiced a moment earlier. An OT case inside an admission cannot receive credit directly; medicine goes to the admission.
Waiting for payment	Admissions that are INVOICED and OT cases that are READY_FOR_DISCHARGE are shown as ready to settle.
Mark paid	When the money reaches the store counter (from the family, or handed over by the operator), the store picks the method. A Payment is created for each unsettled sale (net of credit returns) and the sales become SETTLED. It counts as store income and in **that day's** closing.
Discharge link	The operator cannot discharge the admission or OT case until the store has marked it paid.

10.5 Returns

Sale returns — `SRET-YYYYMMDD-0001`

Find the sale; only quantities not yet returned are offered. Choose **Return to sellable stock** (batch restocked) or **Quarantine/Damaged** (logged, stock unchanged). Reason required.

Refund = each item's share of what was actually paid (discounts are pro-rated). Rounding across partial returns can never refund more than was paid.

CASH refund normally. On an **unsettled credit** sale the return is **CREDIT**: it only lowers what the patient owes, and no cash leaves the drawer.

Purchase returns — `PRET-YYYYMMDD-0001`

Pick the supplier; the screen shows every batch from that supplier with remaining stock. Enter return quantities (not above remaining) and a reason. Credited at the batch's purchase cost.

CASH: the supplier refunds money, posted as a negative **SUPPLIER_REFUND** expense.
CREDIT: deducted from the supplier balance; no cash moves.

10.6 Stock control

Function	Behaviour
Stock adjustments	Remove stock from a specific batch for Damage, Expired, Return to supplier or Stock correction, with a required reason. Adjustments **only reduce** stock. The loss is posted as a **MEDICAL_STORE_WASTAGE** expense at cost.
Stock count	A count sheet per batch. Only lines where the counted figure differs are saved. Shortages become ADJUSTMENT_OUT (posted as wastage at cost) and surpluses ADJUSTMENT_IN (not treated as income). If stock moved since the sheet was loaded, the count is refused for those batches and must be redone.
Stock history	The complete ledger for a medicine: every movement with type, quantity, before/after quantity, source document number and user.
Expiry tracker	Batches by days to expiry (negative = expired). "Expiring soon" means within one calendar month. Expired stock can be written off from here.

Stock movement types recorded in the ledger: `OPENING_STOCK` `PURCHASE` `SALE` `SALE_RETURN` `PURCHASE_RETURN` `ADJUSTMENT_IN` `ADJUSTMENT_OUT` . Stock can never go negative: sales, adjustments and purchase returns are all checked against the batch's remaining quantity.

10.7 Day closing — `/medical-store/day-closing`

Cash sales
All cash payments on store sales that day, including in-patient bills marked paid in cash.

−

Cash refunds
Sale returns refunded in cash that day. Credit returns are excluded.

=

Expected cash
What should be in the drawer.

vs

Counted cash
Entered by staff; difference and note saved.

Non-cash methods are shown separately for information. Figures are always recomputed on the server at save time. One closing per day; re-closing the same day overwrites it. A future day cannot be closed.

10.8 Store dashboard and reports

- **Dashboard** (refreshes every 30 s): today's revenue, low-stock, expiring-soon and expired counts; revenue trend (today/week/month/year); stock-status pie in which each medicine is in exactly one bucket (worst status wins); low, expiring and expired panels. Medicines never purchased are not counted as low stock.
- **Reports** (date range, Excel export): sales revenue, net revenue after returns, purchases cost, purchase returns, current stock value at cost, discounts, profit, top-selling medicines, sales by category, purchases by supplier.

> **Store revenue recognition**
> Counter sales count on the day sold; cash refunds come off on the day refunded. In-patient credit counts only once **SETTLED**, dated on the settlement date, minus any credit returns made before payment. Profit uses the cost recorded on each sale line; older lines without a recorded cost are excluded from profit entirely rather than shown as 100% margin.

11 General inventory (OT stock) — /inventory

Separate from the pharmacy, management keeps a simple register of hospital items, mainly for the operation theatre.

- Item fields: name, category (**MEDICINE**, **TOOL**, **CONSUMABLE**), unit, quantity on hand, low-stock level, unit price. Name + category must be unique.
- **Opening stock** and every **restock** are real purchases, so each is also posted as an **INVENTORY** expense (the cost defaults to quantity × unit price and can be overridden), with a payment method.
- Every movement is kept as an inventory transaction with user and reason.

> **Two separate stock systems**
>
> The medical store (Section 10) has batch/expiry tracking and its own sales. General inventory is a quantity register for OT tools and consumables. Neither updates the other.

12 Administrative and financial management

12.1 The payment ledger

Every amount received is a **Payment** with an amount, method (CASH, CARD, BANK_TRANSFER, ONLINE, FREE), status (PAID, REFUNDED, PARTIALLY_REFUNDED, CANCELLED), a note, the user who took it and the time. Each payment is linked to **exactly one source**:

Source	Reported as	Typical notes
OPD visit	OPD consultation	FREE visits: PKR 0 with the waiver reason
Diagnostic receipt	Laboratory / X-Ray / Ultrasound / ECG / ECO	Counter receipts only (admission tests are paid through the admission)
OT case	Operation theater	Admission payment, advance, part-payment, added item
Emergency visit	Minor emergency	—
Medicine sale	Medical store	Counter sale, or in-patient medicine when marked paid
Admission	In-patient admission	Advance / part-payment / final payment; a negative row for an excess handed back
(none found)	Other / unlinked	Still real money, kept in totals

12.2 When income is recognised

Event	Counted as income on
OPD, emergency, counter tests, counter medicine	The moment of payment
OT and admission advances and instalments	Each instalment on the day it is received
Excess advance handed back	Negative payment on the day it is handed back
In-patient medicine on credit	The day the store marks it paid
Cancelled / refunded test receipt	Drops out (payment status changes)
Medicine cash refund	Deducted from gross and store income on the refund day

12.3 Doctor earnings model

Doctor money is collected by the hospital but **belongs to the doctor**. It is not hospital income, and paying it out is not a hospital expense: it settles a debt. Each doctor's share is computed from four streams:

Stream	Collected	Doctor's share
OPD consultations	Consultation fee per visit, dated by visit date	Hospital split: **PERCENTAGE** = hospital keeps X% of the total; **FIXED** = hospital keeps PKR X per visit/case (capped at the amount collected). Doctor gets the rest.
OT surgeon fee	Doctor fee on **discharged** OT cases, dated by discharge	

Stream	Collected	Doctor's share
Admission visits	Daily doctor fee on **visited** days of **discharged** admissions, dated by discharge	Separate admission split: PERCENTAGE of each day's fee, or FIXED PKR per day kept by the hospital (capped at that day's fee).
Diagnostic referral	PAID receipts where the doctor is the referrer, dated by receipt	Per-module share %: Laboratory, X-Ray, Ultrasound, ECG, ECO (each 0–100%) of the receipt total.

Worked example

Dr. A: consultation fee PKR 1,000, hospital split 30% (percentage), lab share 10%. In a week: 20 OPD visits (PKR 20,000) and lab receipts of PKR 15,000 referred.
Hospital keeps 30% × 20,000 = 6,000; doctor's OPD share = 14,000. Lab referral = 10% × 15,000 = 1,500. **Doctor share = 15,500**, owed by the hospital until paid out.

12.4 Doctor payouts — `/payouts`

- The doctor table shows, per doctor, total collected, hospital share, diagnostic share and doctor net share **for the selected period** (today/week/month/custom).
- **Paid to date** and **Balance due** are **lifetime** figures (lifetime earned − lifetime paid), so switching the period never changes what is actually owed.
- A payout is recorded as a **DOCTOR_PAYOUT** entry linked to the doctor (amount, method, date, note). A helper pre-fills the balance.

12.5 Employees and salaries — `/employees`, `/payouts`

- Employee: name, designation, contact (mobile format), joining date, monthly salary, **payment cycle day** (1–31), active flag.
- **Payday:** the cycle day each month, clamped to the month's last day (e.g. day 31 in February → 28th/29th).
- **First pay is prorated** if the employee joined part-way through a cycle: salary × days worked ÷ days in the cycle. Joining exactly on a payday is not a completed cycle; the first payday is the next one.
- **Status:** Not set · Upcoming · Due today · Overdue · Paid. The salary-due list shows every unpaid employee whatever the payday.
- **Paying salary:** recorded as a **SALARY** expense with optional breakdown base + incentive − deduction (with notes). The server recomputes the total from the parts. A salary **cannot be paid before its payday**, and a payday already settled cannot be paid again.
- A payment settles the most recent payday on or before the payment date. An early small payment (an advance) does not settle the first payday unless it covers the full amount owed.

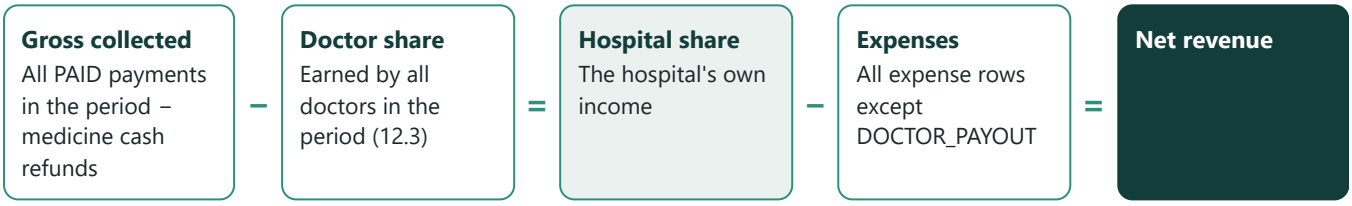
12.6 Expenses — `/expenses`

Category	Entered by	Meaning
GENERAL, LABORATORY, X-RAY, ULTRASOUND, ECG, ECO, OT	Management (manual)	Running costs by department
SALARY	Management (Payouts)	Employee salary, linked to the employee

Category	Entered by	Meaning
DOCTOR_PAYOUT	Management (Payouts)	Settles the doctor balance. **Excluded** from expenses and net revenue
INVENTORY	System (General inventory)	Opening stock and restocks
SUPPLIER_PAYMENT	System (Store)	Money paid to medicine suppliers
MEDICAL_STORE_WASTAGE	System (Store)	Damaged, expired or missing stock at cost
SUPPLIER_REFUND	System (Store)	Negative expense: cash refunded by a supplier

Every expense has an amount (> 0, except the system's negative supplier refunds), method, date and note. The ledger is filtered by date range.

12.7 Net revenue



Cash basis for medicine purchases

Medicine purchases do not hit expenses when stock is received. They hit expenses when the supplier is paid (SUPPLIER_PAYMENT). Unpaid supplier balances are therefore not in net revenue until settled.

12.8 Management dashboard and reports

Dashboard — `/`

Period picker (today, week-to-date, month-to-date, custom) with comparison to the previous equivalent period. Gross collected, doctor share, hospital share, expenses, net revenue; revenue by service; revenue trend (one year); how each rupee was split; doctor accounts (earned/paid/owed, all time); in-patients and OT now; patients seen; registered patients, active doctors, lab receipts and OT cases (all time); store wastage and supplier refunds; latest payments. Includes a refresh button.

Billing & reports — `/reports`

Date range; KPI cards; income by service; full transaction detail including medicine refunds as negative rows; expenses; doctor share by doctor; doctor payouts. **Excel export** (Summary, Income by service, Transactions, Expenses sheets) and a **printable PDF**.

13 Configuration and master data

Master data	Owner	Fields and rules
Hospital settings	Management	Name and address in English and Urdu, phone, email, logo (image upload). Name and logo also appear on the login screen and on printouts.
Doctors	Management	Name, speciality and qualifications in English and Urdu; weekly availability (days + from/to time); consultation fee; OPD hospital split (percentage ≤ 100 or fixed PKR); diagnostic share % per module (0–100); admission daily fee and admission split. Doctors are deactivated, not deleted.
Rooms & beds	Management	Unique name, room type, daily rate, category (Private, Ward, Critical, NICU), unit type (per room / per bed), active. An occupied bed cannot be deactivated.
Admission technician fees	Management (operators read)	Default technician fee per day for each admission category. New admissions copy it; existing stays keep theirs.
Test & price catalog	LAB	Module, name, price, default remarks, result template (Section 9.2).
Medicine catalog, suppliers	Medical store	Section 10.
Employees	Management	Section 12.5.
General inventory	Management	Section 11.
User accounts	Management	Name, username, role, active; password reset (Section 4.4).

First-day setup checklist (from the deployment guide)

1. Change the owner's password.
2. Hospital settings: name, address, phone, logo.
3. Doctors: fee, hospital/doctor split, availability, diagnostic shares, admission fee and split.
4. Test & price catalog (LAB login): lab, X-Ray, ECG, ECO and ultrasound tests and prices.
5. Rooms & beds with technician fees; Employees with payday; Inventory.
6. Medicine catalog (pack levels such as BOX, CARTON, never the base unit) and Suppliers.
7. A login for every staff member.
8. Print one OPD slip and one receipt on the real thermal printer.

14 Business rules catalogue

The rules below are enforced by the server and cannot be bypassed from the screen.

ID	Area	Rule
BR-01	Patients	Name, father's name and gender are required; CNIC and mobile, when given, must be validly formatted and unique.
BR-02	Patients	MR number = `MR-<year>-<6-digit id>`, assigned once, never changed.
BR-03	OPD	Tokens restart at 1 per doctor per day and are never duplicated, even under simultaneous issue.
BR-04	OPD	A FREE consultation requires a waiver reason and is recorded at PKR 0.
BR-05	OPD	An OPD visit can be deleted only with a reason and only if no test receipts were issued under it.
BR-06	Billing	Any discount requires a reason and cannot exceed the subtotal (tests and medicine).
BR-07	Billing	Cancelling or refunding a test receipt requires a reason; its payment follows the receipt's status.
BR-08	Lab	Catalog prices and names are copied onto receipts; result template rows are copied into each report.
BR-09	Lab	The LAB role may enter results only for Laboratory-module tests.
BR-10	Lab	Numeric results must be numbers; qualitative results must be Positive or Negative. Flags are computed, not typed.
BR-11	OT	A payment method (or Due) must be chosen explicitly when creating a case or adding an item; no payment may exceed the balance.
BR-12	OT	Discharge requires zero balance and no unpaid store medicine.
BR-13	OT	An OT case linked to an admission takes no payments or discharge on the OT page; extra items only as Due.
BR-14	Admissions	One active admission per patient; one active admission per configured bed.
BR-15	Admissions	Bed category must match the admission category (Medical/Surgical → Private/Ward; Critical → Critical; NICU → NICU); manual rooms always allowed.
BR-16	Admissions	Stay < 24 h = 1 day; otherwise every calendar date is charged. Each date is charged once.
BR-17	Admissions	A day marked "doctor did not arrive" carries no doctor fee and no doctor earnings.
BR-18	Admissions	Changes to charges, tests, operations, rates and doctor visits are allowed only while ADMITTED.
BR-19	Admissions	After invoicing, collections may not exceed the balance; refunds of excess only after invoicing.
BR-20	Admissions	Discharge requires INVOICED status, hospital balance exactly 0 and no store credit owed.
BR-21	Admissions	Cancellation requires a reason and no payments, medicines, active operations or active tests.
BR-22	Store	Sales draw First-Expiry-First-Out; expired batches are never sold; stock never goes negative.
BR-23	Store	In-patient credit can be issued only while the admission is ADMITTED or the OT case is open; no discount or payment on credit sales.

ID	Area	Rule
BR-24	Store	Credit sales become store income only when marked paid (dated on settlement).
BR-25	Store	Returns on unsettled credit sales are CREDIT (no cash); otherwise CASH at the pro-rated paid price.
BR-26	Store	Adjustments only reduce stock and are posted as wastage at cost; stock counts are refused if stock moved meanwhile.
BR-27	Store	Packaging level names must be unique and differ from the base unit.
BR-28	Store	Day closing is recomputed on the server; one per day; future days cannot be closed.
BR-29	Finance	Each payment belongs to exactly one source.
BR-30	Finance	Doctor payouts are excluded from expenses and net revenue (the doctor share is already deducted from gross).
BR-31	Finance	Doctor balance due is always lifetime earned − lifetime paid, independent of the period filter.
BR-32	Payroll	Salary cannot be paid before its payday or twice for the same payday; first pay is prorated.
BR-33	Users	Users are never deleted; deactivation ends sessions; new/reset accounts must change password first.
BR-34	Security	5 failed logins lock a username for 15 minutes; sessions last 8 hours.
BR-35	Audit	Every create, update, cancel, discount and refund writes an audit log entry with user, time and details.

15 Data model

The database has about 40 tables. Table and column names are snake_case. Money uses fixed decimals. The main entity groups:

Group	Entities	Notes
People & access	User, Session, AuditLog, HospitalSettings	Sessions store only token hashes; one settings row.
Patients & encounters	Patient, OpdVisit, EmergencyVisit	Unique (doctor, date, token) for OPD; unique (date, token) for emergency.
Doctors & staff	Doctor, Employee	Doctor holds fee, splits and module shares; Employee holds salary and cycle day.
Diagnostics	DiagnosticCatalogItem, DiagnosticTestParameter, DiagnosticReceipt, DiagnosticReceiptItem, DiagnosticResultValue	Receipt links to an OPD visit *or* an admission.
OT	OtCase, OtLineItem	Optional link to an admission and to a configured bed.
Admissions	Admission, AdmissionDay, AdmissionCharge, AdmissionCategoryFee, RoomBed	Unique (admission, charge date) guarantees one charge per day.
Money	Payment, Expense, DailySequence	Payment has six optional source links (exactly one used).
General inventory	InventoryItem, InventoryTransaction	OT tools and consumables.
Medical store	MedicineItem, MedicinePackagingLevel, Supplier, SupplierPayment, MedicinePurchase, MedicineBatch, MedicineSale, MedicineSaleItem, MedicineSaleReturn(+Item), MedicinePurchaseReturn(+Item), MedicineStockAdjustment, MedicineStockTransaction, MedicineDayClosing	Batch = unit of stock and expiry; every movement in MedicineStockTransaction with before/after quantities.

15.1 Key relationships

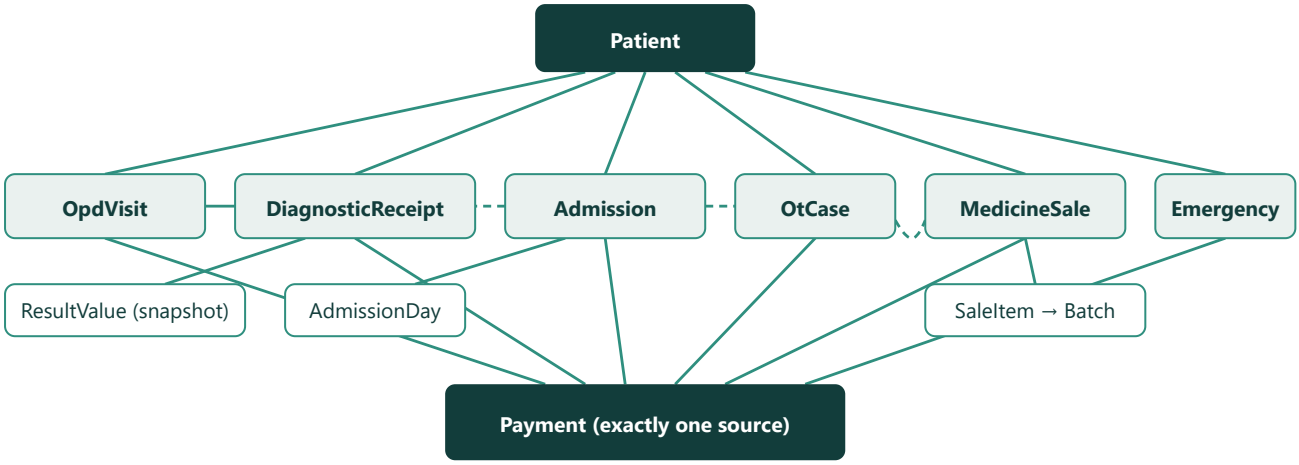


Figure 15.1 — Core relationships. Every encounter belongs to one patient; every payment belongs to one encounter.

15.2 Status values

Field	Values
Payment / receipt / sale status	PAID, REFUNDED, PARTIALLY_REFUNDED, CANCELLED
Payment method	CASH, CARD, BANK_TRANSFER, ONLINE, FREE (OPD waiver only)
OT status	BOOKED, IN_PROGRESS, READY_FOR_DISCHARGE, DISCHARGED, CANCELLED
Admission status	ADMITTED, INVOICED, DISCHARGED, CANCELLED
Medicine sale credit status	(null for counter sales), UNSETTLED, SETTLED
Medicine purpose	WARD, OT, HOME
Return refund mode	CASH, CREDIT
Audit actions	CREATE, UPDATE, CANCEL, DISCOUNT, REFUND, LOGIN, LOGOUT

16 Audit, security and data integrity

16.1 Audit trail

Every change writes an audit log entry: action, entity, entity id, user, time, an optional reason, and before/after details (JSON). Patient edits store both the old and the new values, and the patient history screen shows them. Deleted OPD visits keep their full content in the log, including which slips were returned.

16.2 Integrity safeguards

Safeguard	How
Atomic saves	Multi-step operations (a sale with batches and stock ledger, an admission with its advance, a settlement) run in one database transaction. Either everything is saved or nothing is.
Concurrency-safe numbering	Daily-sequence table plus unique keys; the transaction is retried automatically on a duplicate-key or deadlock clash. OPD and emergency also self-heal a stale sequence.
Race re-checks	Critical conditions are re-checked inside the transaction (e.g. "admission just invoiced" before issuing store medicine; "stock moved" during a count).
Date handling	Date-only columns (visit, sale, expense dates) are compared on correctly anchored day boundaries so "today" in Pakistan never shifts by a day.
Server-side validation	All amounts, formats and status transitions are validated on the server with plain-language error messages for staff.
Security headers	Set in the application configuration; HTTPS enforced; session cookie is secure in production.
Credentials	No real passwords in source files; seed and developer passwords come from the server's environment file.

16.3 Backup and recovery

- Production: cPanel full backups and phpMyAdmin exports before every update and at least weekly, kept off-server, with a monthly test restore.
- Developer console (when enabled): database backup and restore (with an automatic safety backup before restoring), database wipe with typed confirmation, and creation or reset of a management login for lock-out recovery. Its own log is kept outside the database.

17 Deployment and operations

Topic	Details
Production host	GoDaddy cPanel "Setup Node.js App", Node.js 20+, InnoDB database, free AutoSSL certificate with forced HTTPS.
Initial install	Create database and user → import `deploy/schema.sql` in phpMyAdmin → create the owner login → upload app → Run NPM install → `npx prisma generate` and `npm run build` → Restart → check `/api/health` returns `{"status":"ok"}`.
Updates	Export the database first → upload changed files (never overwrite `.env`) → npm install if dependencies changed → generate and build → import any new SQL → Restart.
Schema changes	Must be applied in four places: the Prisma schema, a developer migration, the idempotent start-up update script (adds missing structures without touching data), and `deploy/schema.sql` for fresh installs.
Time zone	The server sets `TZ=Asia/Karachi` at start-up if unset, so "today" is the clinic's day.
Health check	`/api/health` confirms the database is reachable.
Configuration	Environment file: database connection and SSL mode, seed passwords, developer console switch and password, backup tool paths, automatic database updates switch.

Printing

Slips, receipts, invoices and lab reports open as print-ready pages sized for **80 mm thermal paper**. The OPD doctor slip is a full-page bilingual layout. User-entered values are escaped before printing. Printable documents: OPD doctor slip and doctor/patient copies, test receipts, lab reports, emergency receipt, OT admission slip and invoices, admission invoice, medicine sale receipt, reorder list, and management and store reports.

18 Observations and recommendations

These points came up while analysing the system. They are not defects in the rules described above; they are areas for management and the development team to consider.

#	Observation	Recommendation
1	No electronic clinical record (prescriptions, vitals, nursing/progress notes, discharge summary).	If clinical documentation is a goal, add an OPD prescription/notes form and an admission discharge summary linked to the existing visit and admission records.
2	No automated test suite; correctness relies on type-checking and manual checks.	Add automated tests around the money rules first: admission bill and day accrual, doctor share, salary cycle, FEFO and returns.
3	Login lock-outs are kept in memory and reset when the server restarts.	Acceptable for a single process; persist them if the app ever runs on more than one process.
4	A standalone OT case can show status CANCELLED, but no screen action cancels one.	Add a "cancel case" action with a reason and the same "nothing billed or paid" guard as admissions.
5	Medicine purchases are expensed only when suppliers are paid (cash basis).	Management should read net revenue alongside the supplier balances; consider an accrual view in reports.
6	OPD visit deletion removes the visit and payment rows (kept only in the audit log).	Consider a "cancelled" status instead of deletion, for easier reconciliation.
7	Doctor availability is advisory only; slips can be issued outside schedule.	Intended flexibility; optionally warn the operator.
8	Legacy items remain: PDF-upload result columns, typed OT medicine fees on old cases, and two unused libraries.	Keep the data for history; remove unused libraries in a maintenance release.
9	The developer console can wipe or restore the database.	Keep it disabled in production except during supervised maintenance.
10	Backups depend on staff following the routine.	Schedule cPanel backups and keep a written backup log with the monthly restore test.

A Document numbering formats

Document	Format	Sequence resets
Patient MR number	`MR-2026-000412`	Never (database id)
OPD visit	`OPD-20261021-007-0015` (date, doctor id, token)	Daily, per doctor
Diagnostic receipt	`LAB-20261021-0003` · `X-R-…` · `ULT-…` · `ECG-…` · `ECO-…`	Daily, per module
Emergency visit	`EMG-20261021-0002`	Daily
OT case	`OT-20261021-0001`	Daily
Admission	`ADM-20261021-0001`	Daily
Medicine sale	`MED-20261021-0042` (+ daily token)	Daily
Medicine purchase	`PUR-20261021-0001`	Daily
Sale return	`SRET-20261021-0001`	Daily
Purchase return	`PRET-20261021-0001`	Daily

B API reference

All endpoints are under `/api`, return JSON, require a signed-in session (except health, login and the public branding read) and check the role listed.

Endpoint	Methods	Roles / purpose
auth/login · logout · me	POST · POST · GET	Sign in/out, current user
auth/change-password · profile	POST · PATCH	Any signed-in user
patients · patients/[id]	GET, POST · GET, PUT	Operator, Mgmt (Store: read)
opd	GET, POST, DELETE	Operator, Mgmt
emergency	GET, POST	Operator, Mgmt
diagnostics	GET, POST	Bill tests (Operator, Mgmt); lookups (+Lab)
diagnostics/[id]/cancel · print	PATCH	Operator, Mgmt
diagnostics/[id]/results · result	GET, PATCH	Lab (Laboratory only), Mgmt
diagnostics/results · recent-results	GET	Operator, Mgmt, Lab
catalog · catalog/[id]	GET, POST · PATCH	Lab
ot · ot/[id] · ot/all	GET, POST · GET, PATCH · GET	Operator, Mgmt
admissions · admissions/[id]	GET, POST · GET, PATCH (13 actions)	Operator, Mgmt
admission-fees	GET · PUT	Operator+Mgmt read · Mgmt write
rooms · rooms/[id]	GET, POST · PATCH	Mgmt
doctors · doctors/[id]	GET, POST · PATCH	Mgmt
employees · employees/[id]	GET, POST · PATCH	Mgmt
finance	GET, POST	Mgmt: payouts, expenses, salary
inventory · inventory/[id]	GET, POST · PATCH	Mgmt
dashboard · dashboard/operator	GET	Mgmt · Operator, Mgmt
reports/export	GET	Mgmt (Excel)
users · users/[id]	GET, POST · PATCH	Mgmt
hospital-settings	GET · PUT	Public read (branding) · Mgmt write
medical-store/medicines (+[id], [id]/batches)	GET, POST, PATCH	Store, Mgmt
medical-store/suppliers (+[id]) · supplier-payments	GET, POST, PATCH	Store, Mgmt
medical-store/purchases · purchase-returns	GET, POST	Store, Mgmt
medical-store/sales · sale-returns	GET, POST	Store, Mgmt
medical-store/inpatients	GET, POST (mark paid)	Store, Mgmt

Endpoint	Methods	Roles / purpose
medical-store/adjustments · stock-count	GET, POST	Store, Mgmt
medical-store/stock-history · expiry · reorder	GET	Store, Mgmt
medical-store/day-closing	GET, POST	Store, Mgmt
medical-store/dashboard · reports · reports/export	GET	Store, Mgmt
dev/* (login, status, backup(s), restore, clean, create-admin)	various	Developer console only (off in production)
health	GET	Public: database reachability

C Glossary

Term	Meaning
MR number	Medical Record number: the patient's permanent identifier.
OPD	Out-Patient Department: a walk-in consultation with a doctor.
Token	The patient's queue number for a doctor (or the store/emergency) on that day.
OT	Operation Theatre.
NICU	Neonatal Intensive Care Unit.
Admission day	One charged calendar date of an in-patient stay (room + doctor visit + technician).
Invoice (admission)	The final bill; daily charges stop at that moment.
Hospital split	The part of a doctor's fee the hospital keeps (percentage or fixed PKR).
Diagnostic share	The referring doctor's commission % on tests, per module.
Base unit	The smallest unit a medicine is stocked and sold in (e.g. a strip or a vial).
Packaging level	A pack size defined relative to the level below (strip, box, carton…).
Batch	One purchase line of a medicine with its own expiry, cost and remaining quantity.
FEFO	First-Expiry-First-Out: sell the soonest-expiring stock first.
Credit sale	Medicine issued to an in-patient or OT case without payment, settled later.
Settle / mark paid	The store records payment of a patient's credit medicine.
CASH / CREDIT return	A return that hands money back, versus one that only lowers what is owed.
Wastage	Medicine written off (damaged, expired, missing), expensed at cost.
Day closing	The store's end-of-day cash count against expected cash.
Gross / hospital share / net revenue	All money collected / after the doctors' share / after expenses.

End of document.